

SEMINARIUM 3: SAMMANFATTNING ENVARIABELS ANALYS

$$\text{Ex: } \int x^3 e^{x^2} dx = \left[\begin{array}{l} t = x^2 = \\ 2x dx = dt \\ x dx = \frac{1}{2} dt \end{array} \right]$$
$$= \int (x^2)' e^{x^2} dx =$$

$$= \frac{1}{2} \int t e^t dt = \left[\begin{array}{ll} u = t & u' = 1 \\ v = e^t & v' = e^t \end{array} \right]$$

$$= t e^t - \int e^t \cdot 1 dt = t e^t - e^t + C$$

$$\frac{1}{2} (x^2 e^{x^2} - e^{x^2}) + C = \frac{1}{2} e^{x^2} (x^2 - 1) + C$$

EX:

$$\int x^2 \sin(x) dx = \frac{2}{3} x^3 - \frac{2}{3} x^2 + \frac{2}{3} x - \frac{2}{3}$$

$$\begin{bmatrix} u = x^2 & u' = 2x \\ v = -\cos(x) & v' = \sin(x) \end{bmatrix} =$$

$$= \int x^2 (-\cos(x)) dx = -\int x^2 \cos(x) dx$$

$$= \int x^2 \cos(x) dx =$$

$$\int x^2 \sin(x) dx = x^2 - \frac{2}{3} x^3 + \frac{2}{3} x^2 - \frac{2}{3} x + \frac{2}{3}$$

$$= u = \sin(x) \quad u' = \cos(x)$$

$$v = \frac{x^3}{3}$$

$$v =$$

$$\int x^2 \sin(x) dx = \frac{2}{3} x^3 - \frac{2}{3} x^2 + \frac{2}{3} x - \frac{2}{3}$$

$$u = x^2$$

$$u' = 2x$$

$$u' =$$